

Twelve Clebsch graphs and a new upper bound for the Ramsey multiplicity of K_4

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Abstract

The Ramsey multiplicity constant c_4 is the smallest possible limiting density of monochromatic copies of K_4 in a red/blue coloring of K_n . The best construction with public data, due to Parczyk, Pokutta, Spiegel, and Szabó (2022), is a Cayley graph on 768 vertices with density 0.0301449. Their paper also reports an improvement by McKay to 0.0301423 whose graph has not been released, and a 2026 seminar by Feinstein announced a construction below 0.030139 without giving its value. We show that the 768-vertex graph has a simple hidden structure: its vertices split into twelve families, each a fourfold blow-up of the Clebsch graph, and the colors between families follow one of four patterns according to a rule on $\mathbb{Z}_3 \times \mathbb{Z}_2 \times \mathbb{Z}_2$. Making the fractional patterns adjustable gives a two-parameter step graphon on 192 classes whose density is an explicit polynomial; its minimum, 0.03013898, is already below McKay’s value. Refining this graphon with a pentagon pattern and two balanced sign splits gives a 3840-class step graphon with exact density

$$U = \frac{8450462766487926638466333426306607129}{280384030360880691940646801777885184000} = 0.0301388875665\dots,$$

so $c_4 \leq U$. This is below Feinstein’s announced threshold. Both this bound and the bound $c_4 \leq 0.0301389941$ from a rational point of the two-parameter family are proved in Lean, including their numerical evaluations, and U is confirmed by an independent exact computation. The constructions were found autonomously by OpenAI’s GPT-6 Astra model running in the Codex harness, with one agent directing eight copies of itself as subagents.

Keywords. Ramsey multiplicity; Clebsch graph; step graphon; exact computation; formal verification; AI-assisted discovery.

1 Introduction

Color each edge of the complete graph K_n red or blue. How few monochromatic copies of K_4 can there be? Let $m_4(n)$ be the minimum, over all colorings, of the proportion of four-vertex sets that span a monochromatic K_4 , and let $c_4 = \lim_{n \rightarrow \infty} m_4(n)$. Coloring each edge by a fair coin gives $2 \cdot 2^{-6} = 1/32$, so $c_4 \leq 1/32$. For triangles, Goodman showed that the analogous random bound is sharp, $c_3 = 1/4$ [8], and Erdős conjectured that random colorings are optimal for every clique [3]. Thomason disproved this for K_4 and every larger clique [25]. Determining c_4 remains open.

Earlier bounds. Every explicit coloring, or more generally every step graphon (Section 2), gives an upper bound on c_4 . Figure 1(a) shows the constructions of Thomason [25, 26], Franek and Rödl [6], and Even-Zohar and Linial [4], with values as reported in [20]. On the other side, flag algebra computations [22] give $c_4 \geq 0.0296$ [9], improving earlier bounds [7, 23, 15]; Kiem, Pokutta,

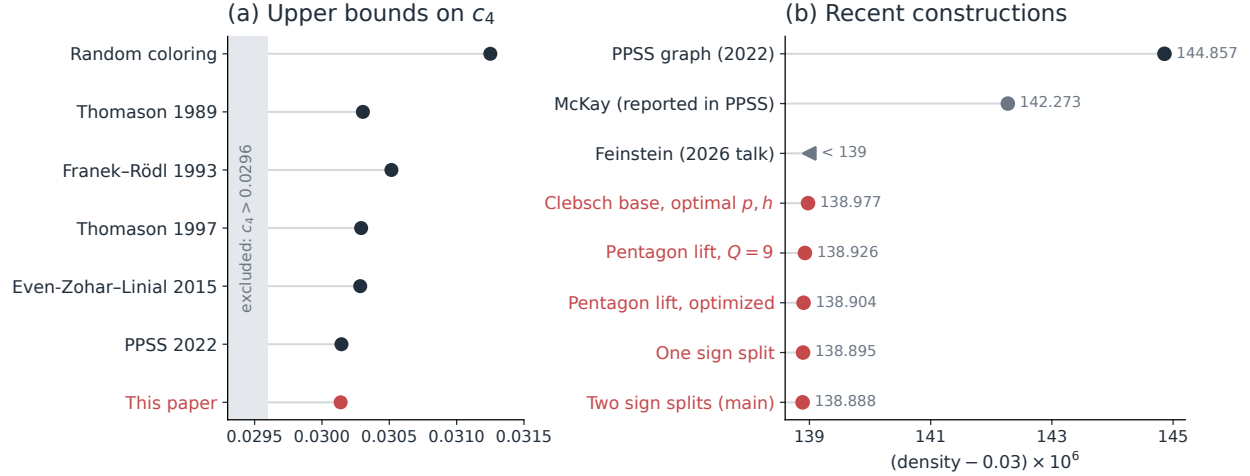


Figure 1: Upper bounds on c_4 . (a) Constructions since Thomason’s disproof of the Erdős conjecture; the shaded region is excluded by the flag algebra lower bound. (b) A zoom on the recent constructions. McKay’s value is reported in [20] without public data, and Feinstein’s announcement gives only the threshold 0.030139. Red points are the constructions of this paper.

and Spiegel report numerical evidence for $c_4 \geq 0.02961$ [12, Section 7]. This paper is about upper bounds only.

The current frontier. Three recent results frame this work (Figure 1(b)).

1. Parczyk, Pokutta, Spiegel, and Szabó (PPSS) searched over Cayley graphs and found a graph on 768 vertices with

$$c_4 \leq \frac{4551721}{150994944} = 0.0301448570 \dots$$

[20, Theorem 1.1]. The paper first appeared in 2022, and its graphs are publicly available [19].

2. A later version of the same paper reports that McKay, starting from the PPSS graph and applying local search, found a 768-vertex graph with density $10486266368/768^4 = 0.0301422734 \dots$ [20, end of Section 5]. That graph has a trivial automorphism group, and we are not aware of a public copy.
3. In January 2026, Feinstein announced randomized constructions showing $c_4 < 0.030139$ in a seminar talk [5]. Neither the exact value nor the construction has been made public, as far as we know.

Our results. We start from the PPSS graph and make three contributions.

Structure. In its published form the PPSS graph is a 768×768 table of colors with little apparent pattern. We show that it is built from the Clebsch graph, the strongly regular graph on $\mathbb{F}_2^4$ with parameters $(16, 5, 0, 2)$. Its vertices split into 12 families of 64, each family into 16 fibers of four vertices, and inside every family the blue edges form the Clebsch graph with each vertex replaced by four. Between two families, the proportion of red edges between fibers follows one of four patterns, and which pattern applies is decided by a rule on the group $\mathbb{Z}_3 \times \mathbb{Z}_2 \times \mathbb{Z}_2$ (Section 3).

A two-parameter family. Replacing the two fractional patterns by free probabilities p and h gives a step graphon $B_{p,h}$ with 192 classes. Its density is an explicit degree-six polynomial whose minimum on $[0, 1]^2$, 0.0301389773, is already below McKay’s value (Section 4).

Refinement. Splitting each class into five with a pentagon pattern, and then twice into two with balanced signs, lowers the density further (Section 5). The result is our main theorem.

Theorem 1.1 (Main bound). *There is a symmetric 3840×3840 table W_* of probabilities with denominator 65536 such that*

$$\mathcal{F}(W_*) = U = \frac{8450462766487926638466333426306607129}{280384030360880691940646801777885184000} = 0.030138887566497220\dots \quad (1)$$

Consequently, for every $n \geq 4$ some red/blue coloring of K_n has monochromatic K_4 proportion at most U , and $c_4 \leq U < 0.030139$.

Theorem 1.1 improves the PPSS bound by $5.97 \cdot 10^{-6}$ and McKay's reported bound by $3.39 \cdot 10^{-6}$. It is also below Feinstein's announced threshold; without an exact value for that construction we cannot say which is smaller, and we make no claim either way. Theorem 1.1, including its numerical evaluation, is proved in Lean, and an independent computation confirms the value (Section 6).

The constructions were found autonomously by an AI research system: a GPT-6 Astra agent in OpenAI's Codex harness directing eight GPT-6 Astra subagents (Section 7).

2 Step graphons and finite colorings

An *equal-class step graphon* on a finite index set I with $|I| = N$ is a symmetric matrix $W = (w_{ij})_{i,j \in I}$ with entries in $[0, 1]$ [14]. We read w_{ij} as the probability that an edge between a vertex of class i and a vertex of class j is red; the diagonal entry w_{ii} describes edges between two distinct vertices of the same class. The *monochromatic K_4 density* of W is

$$\mathcal{F}(W) = \frac{1}{N^4} \sum_{i_1, i_2, i_3, i_4 \in I} \left(\prod_{1 \leq s < t \leq 4} w_{i_s i_t} + \prod_{1 \leq s < t \leq 4} (1 - w_{i_s i_t}) \right), \quad (2)$$

which is $t(K_4, W) + t(K_4, 1 - W)$ in graphon notation. The sum runs over all ordered quadruples, including those with repeated classes.

Lemma 2.1 (Finite realization). *For every step graphon W and every $n \geq 4$, $m_4(n) \leq \mathcal{F}(W)$. Hence $c_4 \leq \mathcal{F}(W)$.*

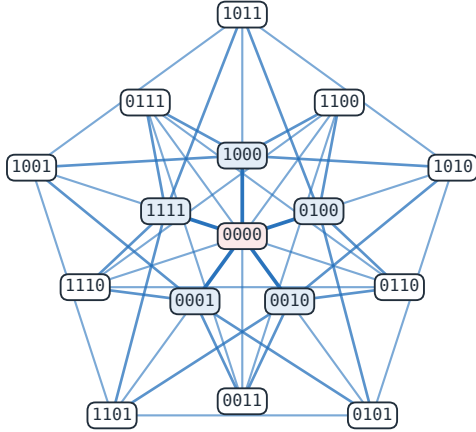
Proof. Give each vertex of K_n an independent uniform class in I , then color each edge red independently with probability w_{ij} , where i and j are the classes of its ends. For four distinct vertices, the four classes are independent and uniform (repetitions allowed) and the six edges are conditionally independent, so the probability that they span a monochromatic K_4 is exactly (2). The expected proportion of monochromatic four-sets is therefore $\mathcal{F}(W)$, and some coloring does at least as well. Finally, $m_4(n)$ is nondecreasing, because averaging over the n -vertex subsets of an optimal coloring of K_{n+1} shows $m_4(n) \leq m_4(n+1)$; so c_4 exists and is at most $\mathcal{F}(W)$. $\square$

When $w_{ij} = r_{ij}/Q$ with integers r_{ij} , put

$$D(r) = \sum_{i_1, i_2, i_3, i_4 \in I} \prod_{s < t} r_{i_s i_t}, \quad \text{so that} \quad \mathcal{F}(W) = \frac{D(r) + D(Q - r)}{N^4 Q^6}. \quad (3)$$

All exact values in this paper are computed this way, in integer arithmetic. A finite graph is the special case of a 0/1 table with zero diagonal, so the PPSS value is $\mathcal{F}$ of its adjacency matrix.

(a) Clebsch graph: $x \sim y$ iff $x + y \in S$



(b) As a coloring of K_{16}

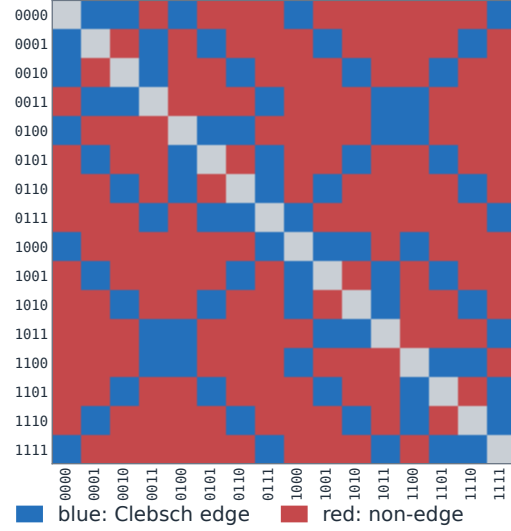


Figure 2: The Clebsch graph on $\mathbb{F}_2^4$. (a) The vertex 0000 is in the center, its five neighbors S around it, and its ten non-neighbors outside. (b) The same graph as a coloring of K_{16} , with Clebsch edges blue. This 16×16 pattern reappears throughout Figures 3 and 4.

3 The Clebsch structure of the PPSS graph

3.1 The Clebsch graph

Let $V = \mathbb{F}_2^4$ with standard basis $e_1, \dots, e_4$, and put

$$S = \{e_1, e_2, e_3, e_4, e_1 + e_2 + e_3 + e_4\}, \quad C = \{0\} \cup S.$$

The Clebsch graph is the Cayley graph on V with connection set S : two binary strings are adjacent when they differ in exactly one coordinate or in all four (Figure 2). It is 5-regular, adjacent vertices have no common neighbor, and nonadjacent vertices have exactly two. In particular it is triangle-free. The five neighbors of any vertex form an independent set, and the ten non-neighbors induce the Petersen graph.

If the Clebsch edges are colored blue and all other pairs red, blue has no triangle and so no K_4 . Red, however, contains many copies of K_4 , so a single Clebsch coloring is far from optimal. The point of the PPSS graph, as we now explain, is how twelve such colorings are coupled.

3.2 Reading the PPSS graph

Let $H = \mathbb{Z}_3 \times \mathbb{Z}_2 \times \mathbb{Z}_2$, whose elements $u = (a, e, s)$ will index twelve *families*. For two families put $u - v = (d_a, d_e, d_s)$ and define the *type*

$$T(u, v) = \begin{cases} Z, & d_s = 0, d_a = 0, d_e = 0 \quad (\text{the same family}), \\ H, & d_s = 0, d_a = 0, d_e = 1, \\ X, & d_s = 0, d_a \neq 0, d_e = 0, \\ P, & d_s = 1, d_a = 0, \\ Z, & \text{otherwise.} \end{cases} \quad (4)$$

Table 1: Red probability between position x of family u and position y of family v , as a function of $z = x + y$ and the type $T(u, v)$. The PPSS column is the proportion of red edges between the corresponding fibers of the PPSS graph; the last column is the two-parameter table $B_{p,h}$. In the PPSS graph the same-position P entry is 1 when $d_e = 0$ and $3/4$ when $d_e = 1$.

Type	PPSS fiber averages			$B_{p,h}$		
	$z = 0$	$z \in S$	$z \notin C$	$z = 0$	$z \in S$	$z \notin C$
Z	0	0	1	0	0	1
X	1	1	0	1	1	0
P	1 or $3/4$	$3/4$	0	p	p	0
H	$1/2$	1	0	h	1	0

The type depends only on $u - v$, and $T(u, v) = T(v, u)$. Each family is of type H with one other family, of type X with two, of type P with two, and of type Z with the remaining six. Figure 4(b) draws this as a graph on the twelve families: for each $a \in \mathbb{Z}_3$ the four families (a, e, s) form a K_4 whose perfect matching $\{d_e = 1, d_s = 0\}$ is of type H and whose remaining four-cycle is of type P , and corresponding families in the three copies are joined by triangles of type X .

Each type comes with a pattern on V , recorded in Table 1 and Figure 4(c)–(f). Pattern Z is the Clebsch coloring itself: blue at the same position and at Clebsch neighbors, red elsewhere. Pattern X is its complement. Patterns P and H are fractional versions of X .

Proposition 3.1 (Structure of the PPSS graph). *The 768 vertices of the PPSS graph [19] can be labeled bijectively by $H \times V \times \{1, 2, 3, 4\}$ so that the following hold, where the fiber (u, x) consists of the four vertices labeled $(u, x, \cdot)$.*

- (i) *Within a family u , an edge between vertices of fibers (u, x) and (u, y) is red if and only if $x + y \notin C$. In particular each fiber spans a blue K_4 , and the blue graph inside a family is the Clebsch graph with each vertex replaced by a blue clique of size four.*
- (ii) *For $u \neq v$, the proportion of red edges among the 16 edges between fibers (u, x) and (v, y) is the PPSS entry of Table 1 for type $T(u, v)$ and $z = x + y$.*

Proof. This is a finite check of all 768^2 entries against an explicit labeling. The labeling is stored with the repository, and the figure generator verifies (i) and (ii) every time it draws Figures 3 and 4. $\square$

Figure 3 shows the effect of the relabeling. In the published vertex order the matrix shows little structure; in the new order it splits into a 12×12 grid of 64×64 blocks, the diagonal blocks are all the same blown-up Clebsch coloring, and every off-diagonal block is one of the four patterns. The relabeling is only a permutation of the vertices, so the density is unchanged.

The two fractional patterns are where the PPSS graph uses finer structure than the Clebsch rule. A $3/4$ block is the complement of a perfect matching between two fibers, and a $1/2$ block is a 2-regular bipartite graph. These 4×4 details matter: a direct count shows that the step graphon of fiber averages has density $9103117/301989888 \approx 0.0301438$, which is slightly smaller than the PPSS value but larger than McKay's. We do not know of earlier work describing the PPSS graph in these terms.

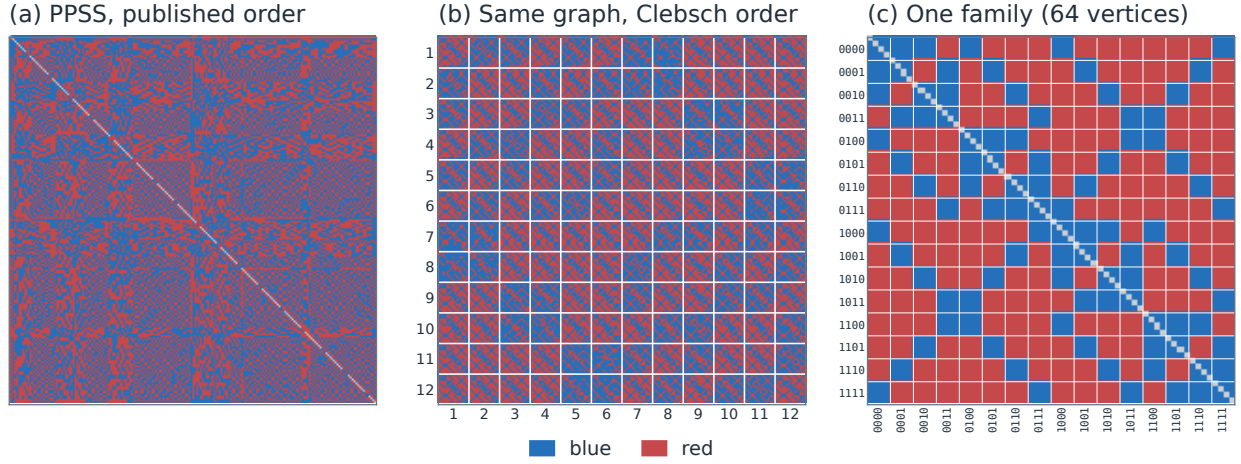


Figure 3: The PPSS graph [19] as a red/blue adjacency matrix. (a) In the published vertex order. (b) The same matrix with the vertices relabeled by Proposition 3.1: twelve families of 64 vertices. (c) One diagonal block: sixteen fibers of four vertices, colored by the Clebsch pattern of Figure 2(b). Gray marks the absent self-pairs.

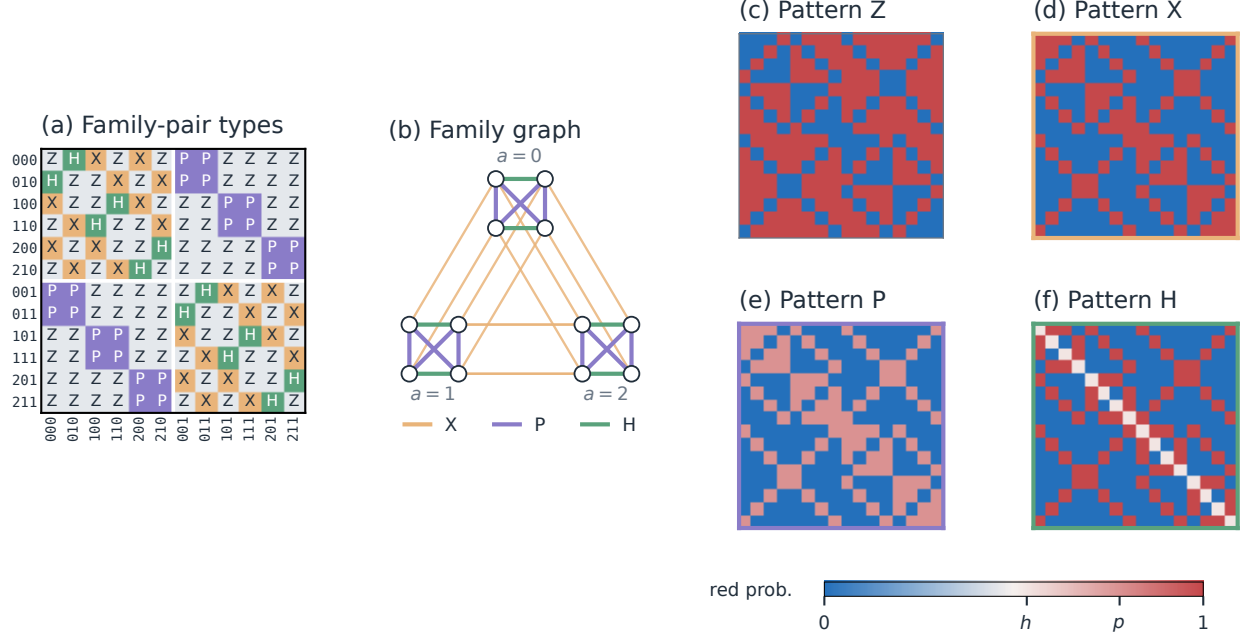


Figure 4: The twelve-family rule. (a) The type $T(u, v)$ for each pair of families, labeled ae s and ordered by s , then a , then e . (b) The same rule as a graph on the families, omitting type Z . (c)–(f) The four 16×16 position patterns, colored at the optimal parameters $p_* \approx 0.779$ and $h_* \approx 0.534$ of Proposition 4.3. Pattern Z is the Clebsch coloring and pattern X its complement.

4 The two-parameter family

Proposition 3.1 suggests replacing every fractional pattern by a free parameter.

Definition 4.1. For $p, h \in [0, 1]$, let $B_{p,h}$ be the step graphon on the 192 classes $I = H \times V$ given by

$$B_{p,h}((u, x), (v, y)) = g_{T(u,v)}(x + y), \quad (5)$$

where $g_Z, g_X, g_P, g_H: V \rightarrow [0, 1]$ are the $B_{p,h}$ columns of Table 1.

The table is symmetric, has zero diagonal, and is invariant under translation by the group $H \times V$ of order 192. Each row has 87 entries equal to 1, twelve equal to p , one equal to h , and 92 equal to 0. Compared with the fiber averages of the PPSS graph, the $3/4$ and $1/2$ entries become p and h , and the 96 same-position P pairs that were completely red also become p .

Proposition 4.2 (Density polynomial). $\mathcal{F}(B_{p,h}) = P(p, h)/7077888$, where

$$\begin{aligned} P(p, h) = & 251655 - 97344p + 66888p^2 - 7008p^3 + 1224p^4 \\ & - 6552h + 9216ph - 4176p^2h + 2304p^3h - 144p^4h \\ & + 990h^2 - 576ph^2 + 648p^2h^2 - 288p^3h^2 + 72p^4h^2 \\ & - 16h^3 + 3h^4. \end{aligned} \quad (6)$$

Proof. Each summand of (2) is a polynomial in p and h of total degree at most six. By translation invariance we may fix the first class to be the identity and divide by 192^3 . Exact evaluation on the grid $\{0, 1, \dots, 6\}^2$ then determines the polynomial by interpolation; the supplement records the computation. $\square$

Proposition 4.3 (Minimum of the family). *On $[0, 1]^2$ the polynomial P has a unique minimum, at*

$$p_* = 0.779180833031354\dots, \quad h_* = 0.534265188030699\dots,$$

with value $\mathcal{F}(B_{p_, h_*}) = 0.030138977289665338\dots$. The coordinates p_* and h_* are algebraic of degree seventeen.*

Proof sketch. A Gröbner basis computation reduces the critical-point equations to a single irreducible polynomial of degree seventeen with exactly one real root in $(0, 1)$; the Hessian there is positive definite, and every boundary value is larger. Appendix D gives the details. $\square$

A nearby rational point gives a compact bound with a short formal proof.

Theorem 4.4 (Compact bound). *For $p = 32/41$ and $h = 22/41$,*

$$\mathcal{F}(B_{p,h}) = U_{192} = \frac{1013294255057839}{33620705806123008} = 0.030138994133588287\dots, \quad (7)$$

and therefore $c_4 \leq U_{192}$.

Proof. Substitute into (6) and apply Lemma 2.1. The Lean proof evaluates the density directly, without the interpolation. $\square$

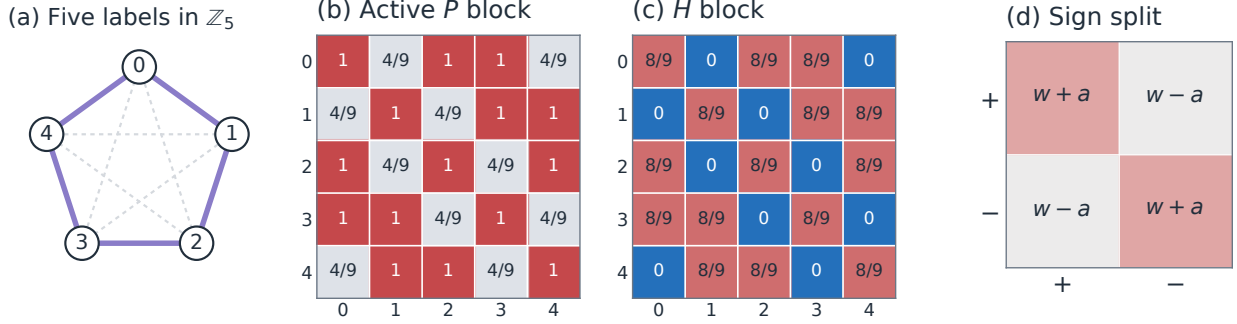


Figure 5: The two refinements. (a) Each class is split into five subclasses indexed by $\mathbb{Z}_5$, and the pentagon $a - b = \pm 1$ marks the *on-pattern* pairs (here with phase $\gamma = 0$). (b) An active P block of $C(9; 7, 4, 8)$: $4/9$ on the pentagon and 1 elsewhere, with every row and column averaging $7/9$. (c) An H block: 0 on the pentagon and $8/9$ elsewhere, averaging $8/15$. (d) A balanced sign split replaces an entry w by a 2×2 block with the same average.

5 Refinements

The minimum of Proposition 4.3 is optimal only for 192 classes. Splitting each class into several equal subclasses, and letting the probabilities vary between subclasses while keeping their block averages, enlarges the family. Because the density depends on products of six entries, a pattern with the same average can have a smaller density. We use two such refinements (Figure 5).

5.1 A pentagon refinement

Replace each base class i by five classes (i, a) , $a \in \mathbb{Z}_5$. Blocks with value 0 or 1, including the diagonal blocks, stay constant. Each of the 1248 fractional pairs $i < j$ (1152 of type P and 96 of type H) receives a phase $\gamma_{ij} \in \mathbb{Z}_5$, or is marked inactive; we set $\gamma_{ji} = -\gamma_{ij}$. A pair of subclasses $(i, a), (j, b)$ is on-pattern if $a - b - \gamma_{ij} \equiv \pm 1 \pmod{5}$.

For integers Q, p_0, x, y , the 960-class table $C(Q; p_0, x, y)$ is:

- on an inactive P pair, constant p_0/Q ;
- on an active P pair, x/Q on-pattern and 1 otherwise;
- on an H pair, 0 on-pattern and y/Q otherwise.

In the phase table we use, 272 of the P pairs are inactive and the other 880 are active. The table is given as finite data in the supplement; we do not have a closed-form description of it.

The simplest example uses only the probabilities 0, $4/9$, $7/9$, $8/9$, 1. Two independent generic counters give

$$\mathcal{F}(C(9; 7, 4, 8)) = \frac{4198776398959}{139314069504000} = 0.030138925766\dots, \quad (8)$$

which is below the minimum of the two-parameter family. Tuning the parameters, and then the amplitude on individual pairs, gives a 960-class table W_0 with $\mathcal{F}(W_0) = 0.030138903565\dots$ (Appendix B).

5.2 Balanced sign refinements

Let W be a step graphon on I and A a symmetric matrix with $|A_{ij}| \leq \min(w_{ij}, 1 - w_{ij})$. The *balanced sign refinement* of W by A is the step graphon on $I \times \{\pm 1\}$ given by

$$W'((i, s), (j, t)) = w_{ij} + st A_{ij}. \quad (9)$$

The bound on A keeps every entry in $[0, 1]$, and each 2×2 block has average w_{ij} . Signed perturbations of this kind are standard in the study of graphon inequalities [13, 1]; here we apply one to a nonconstant table.

Proposition 5.1 (Sign refinement identity). *With $q_{ij} = 1 - w_{ij}$, we have $\mathcal{F}(W') - \mathcal{F}(W) = T_3(A) + T_4(A)$, where*

$$T_3(A) = \frac{4}{N^4} \sum_{i,j,k,l} A_{ij} A_{jk} A_{kl} (w_{il} w_{jl} w_{kl} - q_{il} q_{jl} q_{kl}), \quad (10)$$

$$T_4(A) = \frac{3}{N^4} \sum_{i,j,k,l} A_{ij} A_{jk} A_{kl} A_{li} (w_{ik} w_{jl} + q_{ik} q_{jl}), \quad (11)$$

and both sums include repeated indices.

Proof. Fix classes i, j, k, l and expand the product of the six red entries (9). The four signs are independent and uniform, so the average of a monomial over the signs vanishes unless every vertex of K_4 has even degree in the corresponding edge set. The nonempty even subgraphs of K_4 are its four triangles and its three four-cycles, which give (10) and (11) for red. For blue every amplitude changes sign, which flips the triangle terms and leaves the cycle terms unchanged. $\square$

In particular, scaling A by ϵ gives $\mathcal{F}(W') = \mathcal{F}(W) + \epsilon^3 T_3(A) + \epsilon^4 T_4(A)$, with no linear or quadratic term. So a direction that lowers the density after refinement can be invisible to any first- or second-order test in the unrefined class space. This is why the refinements can go below the minimum of a family whose Hessian is positive definite.

5.3 The final witness

Applying two successive sign refinements to W_0 , with integer amplitudes over 65536, gives tables with 1920 and 3840 classes; the second is the witness W_* of Theorem 1.1. Equivalently, W_* splits each of the 192 base classes into 20 subclasses (a pentagon label and two sign labels). Every entry is an integer multiple of $1/65536$, and every 20×20 block has the same row and column averages as $B_{51064/65536, 35139/65536}$. The supplement gives all 1248 fractional blocks; Appendix B describes the format. Table 2 lists all the constructions.

6 Verification

Table 3 summarizes how each result is established.

The Lean development defines the density (2) literally and proves Lemma 2.1 in general. For W_* , it embeds the 1248 fractional blocks, checks their symmetry, range, and block averages, and proves that an optimized integer counting formula equals the density. Evaluating that formula gives the numerator 519196432373018212667371525712277942005760 over $65536^6 \cdot 3840^4$, which reduces to (1); Lemma 2.1 then gives Theorem 1.1 for every $n \geq 4$, together with the asymptotic bound. The

Table 2: Constructions of this paper, all with exact densities.

Construction	Classes	Density
PPSS graph [20]	768	0.030144857035...
Fiber averages of the PPSS graph	192	0.030143780841...
$B_{32/41,22/41}$ (Theorem 4.4)	192	0.030138994134...
B_{p_*,h_*} (minimum, irrational)	192	0.030138977290...
$C(9; 7, 4, 8)$	960	0.030138925766...
Optimized pentagon refinement W_0	960	0.030138903565...
First sign refinement	1920	0.030138895264...
Second sign refinement W_* (Theorem 1.1)	3840	0.030138887566...

Table 3: How each result is established.

Result	Formal proof (Lean)	Independent computation
Theorem 1.1 (3840 classes)	Complete; numerical step by native evaluation	Character-identity count in Python and C++ (Appendix C)
Theorem 4.4 (192 classes)	Complete; numerical step by native evaluation	Two generic exact counters
Proposition 3.1 (structure)	—	Entry-by-entry check of the PPSS graph
Proposition 4.3 (minimum)	—	Exact computer algebra

proof of Theorem 4.4 has the same shape. There are no `sorry` placeholders or assumed numerical values.

The general arguments are checked by Lean’s kernel using its standard axioms and `mathlib`. The numerical evaluations, and some structural checks on the embedded data, use native evaluation (`native_decide`), which additionally trusts Lean’s compiler and runtime. The independent count shares no code with the Lean proof and uses a different reduction; it agrees exactly, with the red and blue totals summing to the same numerator.

7 How the construction was found

The results in this paper were found by an autonomous AI research system. Every agent was OpenAI’s GPT-6 Astra model [17], running in OpenAI’s Codex agent harness [16]. A parent agent ran the campaign and directed eight subagents, created with Codex’s native subagent mechanism. Each subagent worked on its own line of attack, such as new constructions, algebraic descriptions, optimization, literature review, and verification. The parent kept a registry of approaches, combined the evidence, reassigned subagents as lines succeeded or failed, and decided which candidates to promote.

All agents followed a hypothesis-driven research skill, a set of standing instructions loaded into the harness. We derived the skill from two public prompts for autonomous mathematical research: OpenAI’s prompt for the Cycle Double Cover conjecture [18] and the Jacobian Conjecture prompt hosted by Aaron Lou [10]. From them it keeps diverse approach families, independent early work, a registry of approaches, concrete outputs, and adversarial review. It adds what a computational search needs: falsifiable predictions, small-instance oracles, persistent experiment records, and independent exact recounts. It drops the prompts’ fixed agent counts and run lengths, and it permits literature searches. The supplement contains the skill and a note on how it adapts the two prompts.

The path to Theorem 1.1 went through the steps of this paper in order. The fiber partition of the PPSS graph and its quotient on 192 classes were found first, and a separate line of work then identified the quotient as the Cayley rule (4). Optimization of p and h , the pentagon refinement, and the sign refinements followed, each improving the previous best. The campaign started on 27 September 2026 at Sundai Hack 142 [24], and W_* was produced and independently recounted within its first day.

The system was designed so that no agent could certify its own result. Every proposed improvement was an explicit rational matrix with a hash and a predicted value, and was accepted only after a separate program had recounted it; the final count of W_* was done by a checker that did not read the search code. Agreement between agents was never treated as evidence. The human authors are responsible for the paper and its claims, which rest on the proofs, the explicit witnesses, and the exact computations described above.

8 Discussion

Most of the improvement over PPSS comes from the twelve-family Clebsch structure itself: two parameters reach 0.0301389773 , and the refinements lower this by a further $9.0 \cdot 10^{-8}$, of which the two sign splits contribute $1.6 \cdot 10^{-8}$. The sign identity explains why such gains are possible: they are of third order in the refinement amplitudes and invisible in the unrefined parameter space. We do not know whether further refinements of the same base continue to help, or whether a different

number of families or a different base graph does better; in limited experiments, removing families made the density worse and adding families did not help.

Clebsch and pentagon structures are extremal in several neighboring problems. Pikhurko and Vaughan show that blow-ups of the complement of the Clebsch graph minimize six- and seven-clique densities among graphs with independence number less than three [21]. Kiem, Pokutta, and Spiegel determine the four-color triangle multiplicity through blow-ups of the two three-color Ramsey colorings of K_{16} , whose color classes are Clebsch graphs [12], and pentagon blow-ups give the three-color triangle multiplicity [2]. None of these results implies anything about the optimality of our construction.

Feinstein describes his constructions as structured and symmetric [5]. It would be interesting to know whether they are related to the Clebsch structure described here. The gap to the lower bound 0.0296 remains about $5 \cdot 10^{-4}$, far larger than any of the improvements above.

Data and code availability

Code and data are at <https://github.com/lukevs/k4-ramsey>. The `paper/` directory contains this manuscript, the compact construction data, and the verification records; Appendix A lists the files. The repository also contains the search programs, the independent counters, the Lean development, and the research records.

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We also thank Parczyk, Pokutta, Spiegel, and Szabó for publishing their construction data, which was the starting point of this work, and the authors of the two research prompts [10, 18].

A Supplement and reproduction

Paths are relative to `paper/` unless they start with `../`.

data/final3840.json.gz

The coarse rule, the fractional block index, and all 1248 blocks of W_* . It contains no density value.

../src/k4_ramsey/paper/witness.py

Checks the compact data (block means, range, sign symmetry) and can write the full integer table for an independent count.

data/final3840-audit.json

The character-count record: red and blue totals, primes, orbit representatives, bounds, and validation results.

data/base-polynomial.json, data/base-minimum.json

Coefficients of P , the elimination polynomials, root intervals, and boundary computations.

data/pentagon-q9.json.gz

The table $C(9; 7, 4, 8)$ of (8), with its phase table, class map, and two independent count records.

data/research-workflow.md, data/workflow-source-principles.md

The research skill followed by all agents, and how it adapts the two source prompts.

data/manifest.json

Hashes of all supplement files.

From the repository root, `just paper-check` validates the supplement and `just paper-figures` regenerates the figures from the published data, rechecking Proposition 3.1. The Lean proofs are in `../lean/K4Ramsey/Constructions/Clebsch192/` and `../lean/K4Ramsey/Constructions/Final3840/`; `just certify-final3840` checks the numerical certificate for Theorem 1.1. The independent counter is `../research/experiments/round5_F5/zk_checker.py` with contraction engine `../research/experiments/round5_F5/k4mix.cpp`.

B The optimized pentagon table and the witness format

The table W_0 . Write the pentagon entries around their block means using the centered function

$$\kappa(d) = \begin{cases} 3, & d \equiv \pm 1, \\ -2, & d \equiv 0, \pm 2, \end{cases} \quad \sum_{d \in \mathbb{Z}_5} \kappa(d) = 0.$$

With $Q = 65536$, an active P pair has numerator $51064 + q_{ij}\kappa(a - b - \gamma_{ij})$, an inactive one has 51064 , and an H pair has $35139 - 11713\kappa(a - b - \gamma_{ij})$. With $q_{ij} = -7236$ for every active pair this is $C(65536; 51064, 29356, 58565)$. The table W_0 changes q_{ij} on 234 of the 880 active pairs, to integers between -7236 and -5685 , and has

$$\mathcal{F}(W_0) = \frac{16900934504649027287619486865996291}{560768060721761383881293603555770368} = 0.030138903565399090 \dots \quad (12)$$

The witness W_* . Order the coarse classes as $(a, e, s, x) \in \mathbb{Z}_3 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{F}_2^4$, encode x as an integer in $\{0, \dots, 15\}$, and put $i = 64a + 32e + 16s + x$. Let $Q = 65536$ and $b_{ij} = Q B_{51064/Q, 35139/Q}(i, j)$. (Here $35139/Q$ is the mean of the refined H blocks; it differs slightly from the best unrefined value $35015/Q$.) For each of the 1248 fractional pairs $i < j$, the supplement gives a 20×20 integer block M_{ij} ; put $M_{ji} = M_{ij}^T$, and let M_{ij} be the constant block b_{ij} when $b_{ij} \in \{0, Q\}$, including on the diagonal. Then

$$W_*((i, \alpha), (j, \beta)) = \frac{M_{ij}(\alpha, \beta)}{Q}, \quad i, j \in \{0, \dots, 191\}, \quad \alpha, \beta \in \{0, \dots, 19\}. \quad (13)$$

Every entry is an integer in $[0, Q]$, and every row and column of M_{ij} sums to $20b_{ij}$. The full matrix that was counted has SHA-256 identifier

05302cbc635e939cc41f4ba019cdcba80199b0b83563100bac0b1a0d9fff1a29;

the compact data use a permutation of the coarse classes, and the supplement checks every entry against that matrix. That matrix orders fine labels as $20i + 4a + 2b + c$ with $a \in \{0, \dots, 4\}$ and $b, c \in \{0, 1\}$; the two sign changes act by the XOR masks 1 and 2 on the fine index.

C An independent count by characters

A direct evaluation of (2) for $N = 3840$ has about $2 \cdot 10^{14}$ terms. This independent count uses the sign symmetry of W_* to reduce it. Fourier methods already appear in the multiplicity analysis of Jagger, Šťovíček, and Thomason [11]; we give the specific identity we need.

Changing either sign label of every class simultaneously leaves W_* unchanged, so $G = \mathbb{F}_2^2$ acts freely with 960 orbits. The checker verifies this on every entry before using it. In general, let $G = \mathbb{F}_2^k$ with $K = |G|$, and suppose the integer red table has the form $r((u, x), (v, y)) = f_{uv}(x + y)$ for $u, v \in J$ and $x, y \in G$, with $|J| = n$, so $N = nK$. For $c \in G$ let $\chi_c(x) = (-1)^{c \cdot x}$ and $\hat{f}_c(u, v) = \sum_{z \in G} f_{uv}(z) \chi_c(z)$. Fourier inversion gives

$$r((u, x), (v, y)) = \frac{1}{K} \sum_{c \in G} \hat{f}_c(u, v) \chi_c(x) \chi_c(y). \quad (14)$$

Proposition C.1 (Character count). *For $c = (c_{12}, c_{13}, c_{14}, c_{23}, c_{24}, c_{34}) \in G^6$ let*

$$S(c) = \sum_{u_1, u_2, u_3, u_4 \in J} \prod_{s < t} \hat{f}_{c_{st}}(u_s, u_t),$$

and let $\mathcal{C} \subseteq G^6$ be the set of labelings whose labels sum to zero at each vertex of K_4 . Then $D(r) = K^{-2} \sum_{c \in \mathcal{C}} S(c)$, and $\mathcal{C} = \{(a + b, a + d, b + d, a, b, d) : a, b, d \in G\}$.

Proof. Substitute (14) into each of the six factors of $D(r)$, which contributes K^{-6} . At vertex s the product of the incident characters is χ of the sum of the incident labels, and summing over $x_s \in G$ gives K if that sum is zero and 0 otherwise. The four vertex sums leave exactly the labelings in $\mathcal{C}$, with a factor K^4 . Solving the vertex conditions with $c_{23} = a$, $c_{24} = b$, $c_{34} = d$ gives the description of $\mathcal{C}$. $\square$

The identity applies to red and blue separately. For $k = 2$ there are 64 labelings in $\mathcal{C}$, in eleven orbits under permutations of the four vertices; since every Fourier block is symmetric, one representative per orbit suffices.

Each $S(c)$ is computed modulo several primes $p < 2^{21}$ and reconstructed by the Chinese remainder theorem; the product of the primes exceeds twice the bound $|S(c)| \leq n^4 \prod_{s < t} \max_{u, v} |\hat{f}_{c_{st}}(u, v)|$. The modular contractions run in C++ with Apple Accelerate matrix multiplication; the code checks that $n(p - 1)^2 < 2^{53}$, so every dot product is exact in double precision. Before the full count, the checker confirms the shape, symmetry, range, sign symmetry, and Fourier reconstruction of the input on all 14745600 entries. It agrees with literal arbitrary-precision counts on 27 small cases and on principal submatrices of orders 480 and 960, and with a full generic count on the 1920-class table. For W_* it gives

$$\begin{aligned} D(r) &= 259147196226119060009588075500947710392320, \\ D(Q - r) &= 260049236146899152657783450211330231613440, \\ N^4 Q^6 &= 17226794825372509712833339501233265704960000, \end{aligned} \quad (15)$$

which by (3) is the value U in (1). The value predicted by the search code was compared with this count only after the count was complete.

D The minimum of the two-parameter family

This appendix completes the proof of Proposition 4.3. A lexicographic Gröbner basis of (P_p, P_h) over $\mathbb{Q}$ has the form $\{h - r(p), g(p)\}$, where $r \in \mathbb{Q}[p]$ has degree sixteen and the primitive polynomial

$$\begin{aligned} g(p) = & 288p^{17} - 4968p^{16} + 48528p^{15} - 326808p^{14} + 1640880p^{13} - 6374994p^{12} \\ & + 19669380p^{11} - 49113990p^{10} + 100709306p^9 - 170524126p^8 \\ & + 238109998p^7 - 272230472p^6 + 252306699p^5 - 183872508p^4 \\ & + 106134988p^3 - 50525006p^2 + 20013641p - 4856020 \end{aligned}$$

is irreducible. A Sturm count shows that g has exactly one real root, which lies in $(0, 1)$, and root isolation puts the corresponding h in $(0, 1)$. Interval evaluation gives $\mathcal{F}(B_{p_*, h_*}) \in (0.03013897728966533, 0.03013897728966535)$, and the Hessian there is positive definite (first diagonal entry above 0.01565, determinant above $2.66 \cdot 10^{-6}$). On the boundary, the edges $p = 0$ and $p = 1$ have no stationary points, the minima on $h = 0$ and $h = 1$ are approximately 0.0301635 and 0.0301574, and every corner value exceeds 0.03015. So the interior critical point is the unique global minimum. The supplement gives r , the elimination polynomial for h_* , and exact root intervals.

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